SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 Cyber & Information Security Practical

PRACTICAL - 2

Practical 1(A)

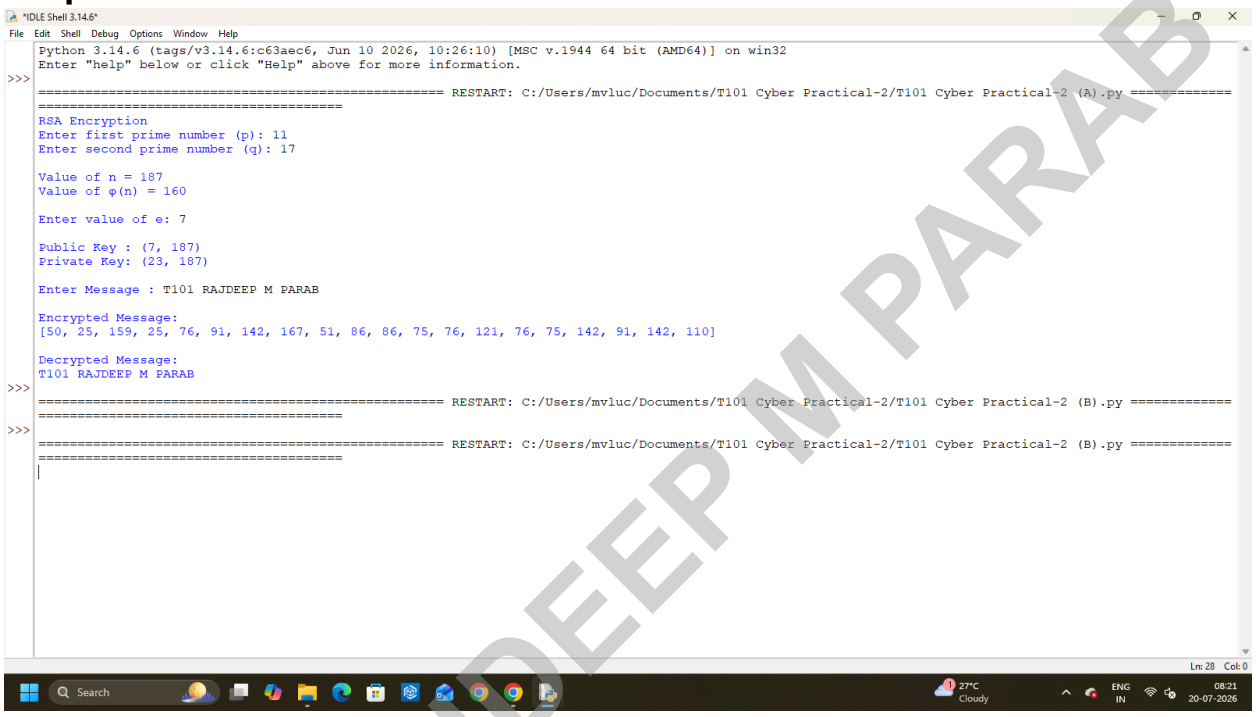
Aim :- Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations. **(CLI)**

Code :-

```
# RSA Encryption and Decryption (CLI)
from math import gcd
# Calculate modular inverse
def mod_inverse(e, phi):
    for d in range(2, phi):
        if (d * e) % phi == 1:
            return d
    return None
# Encrypt message
def encrypt(message, e, n):
    encrypted = []
    for ch in message:
        encrypted.append(pow(ord(ch), e, n))
    return encrypted
# Decrypt message
def decrypt(cipher, d, n):
    text = ""
    for num in cipher:
        text += chr(pow(num, d, n))
    return text
print("RSA Encryption")
p = int(input("Enter first prime number (p): "))
q = int(input("Enter second prime number (q): "))
n = p * q
phi = (p - 1) * (q - 1)
print("\nValue of n =", n)
print("Value of φ(n) =", phi)
while True:
    e = int(input("\nEnter value of e: "))
    if gcd(e, phi) == 1:
        break
    else:
        print("e must be coprime with φ(n). Try again.")
d = mod_inverse(e, phi)
print("\nPublic Key :", (e, n))
print("Private Key:", (d, n))
message = input("\nEnter Message : ")
cipher = encrypt(message, e, n)
print("\nEncrypted Message:")
```

```
print(cipher)
plain = decrypt(cipher, d, n)
print("\nDecrypted Message:")
print(plain)
```

Output :-



SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 Cyber & Information Security Practical

Practical 1(B)

Aim :- Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations. **(GUI)**

Code :-

```
from tkinter import *
from tkinter import messagebox
from math import gcd
# Find modular inverse
def mod_inverse(e, phi):
    for d in range(2, phi):
        if (d * e) % phi == 1:
            return d
    return None
# Encrypt
def encrypt():
    try:
        p = int(entry_p.get())
        q = int(entry_q.get())
        e = int(entry_e.get())
        message = entry_message.get()
        n = p * q
        phi = (p - 1) * (q - 1)
        if gcd(e, phi) != 1:
            messagebox.showerror("Error", "e must be coprime with φ(n)")
            return
        d = mod_inverse(e, phi)
        encrypted = [pow(ord(ch), e, n) for ch in message]
        lbl_public.config(text=f"Public Key : ({e}, {n})")
        lbl_private.config(text=f"Private Key : ({d}, {n})")
        lbl_cipher.config(text="Cipher : " + str(encrypted))
    except:
        messagebox.showerror("Error", "Invalid Input")
# Decrypt
def decrypt():
    try:
        p = int(entry_p.get())
        q = int(entry_q.get())
        e = int(entry_e.get())
        n = p * q
        phi = (p - 1) * (q - 1)
        d = mod_inverse(e, phi)
```

```
        cipher = entry_cipher.get()
        cipher = cipher.split(",")
        cipher = [int(i.strip()) for i in cipher]
        plain = ""
        for num in cipher:
            plain += chr(pow(num, d, n))

        lbl_plain.config(text="Plain Text : " + plain)

    except:
        messagebox.showerror("Error", "Enter cipher correctly.")
root = Tk()
root.title("RSA Encryption & Decryption")
root.geometry("700x620")
root.configure(bg="#E8F4FA")
Label(root,
      text="RSA Encryption & Decryption",
      font=("Arial",18,"bold"),
      bg="#E8F4FA",
      fg="navy").pack(pady=10)
Frame1 = Frame(root,bg="#E8F4FA")
Frame1.pack()
Label(Frame1,text="First Prime Number
(p)",bg="#E8F4FA").grid(row=0,column=0,padx=10,pady=8)
entry_p = Entry(Frame1,width=30)
entry_p.grid(row=0,column=1)
Label(Frame1,text="Second Prime Number
(q)",bg="#E8F4FA").grid(row=1,column=0,padx=10,pady=8)
entry_q = Entry(Frame1,width=30)
entry_q.grid(row=1,column=1)
Label(Frame1,text="Public Exponent
(e)",bg="#E8F4FA").grid(row=2,column=0,padx=10,pady=8)
entry_e = Entry(Frame1,width=30)
entry_e.grid(row=2,column=1)
Label(Frame1,text="Message",bg="#E8F4FA").grid(row=3,column=0,padx=10,pady=8)
entry_message = Entry(Frame1,width=30)
entry_message.grid(row=3,column=1)
Button(root,
       text="Encrypt",
       command=encrypt,
       bg="green",
       fg="white",
       width=18).pack(pady=10)
lbl_public = Label(root,text="",bg="#E8F4FA",font=("Arial",11))
```

```
lbl_public.pack()
lbl_private = Label(root,text="",bg="#E8F4FA",font=("Arial",11))
lbl_private.pack()
lbl_cipher = Label(root,text="",bg="#E8F4FA",font=("Arial",11))
lbl_cipher.pack(pady=10)
Label(root,
      text="Enter Cipher Numbers\n(separate by comma)",
      bg="#E8F4FA").pack()
entry_cipher = Entry(root,width=60)
entry_cipher.pack()
Button(root,
       text="Decrypt",
       command=decrypt,
       bg="blue",
       fg="white",
       width=18).pack(pady=10)
lbl_plain = Label(root,text="",bg="#E8F4FA",font=("Arial",12,"bold"))
lbl_plain.pack(pady=10)
root.mainloop()
```

Output :-